

MID-CAROLINA, NC, USA

WMO: 723156

Lat: 35.646N

Lon: 80.520W

Elev: 773

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.3	-2.9	4.8	30.1	2.4	6.4	32.3	19.4	50.9	17.3	49.8	2.4	10	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	93.1	73.2	90.9	73.2	89.7	72.9	77.6	86.3	76.6	85.2	75.5	83.9	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	136.3	80.0	74.1	131.1	79.2	73.1	126.6	78.5	41.5	86.5	40.5	85.1	39.6	83.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.7	11.8	10.7	95.8	5.3	3.2	6.9	98.1	3.8	99.9	0.8	101.7	-3.0	104.1
			9.9	80.0	4.9	1.6	6.4	81.2	3.5	82.2	0.7	83.1	-2.9	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	40.3	43.3	50.5	59.7	67.2	74.9	78.2	77.2	71.5	60.4	49.2	44.0
	DBStd	15.42	10.34	9.47	9.28	8.03	6.89	4.77	3.85	4.57	5.66	7.94	8.43	8.67
	HDD50	1043	334	229	111	14	0	0	0	0	10	114	231	
	HDD65	3422	768	608	457	195	62	2	0	0	14	187	477	652
	CDD50	4617	32	43	126	305	534	746	873	844	645	333	91	45
	CDD65	1520	1	1	7	36	132	298	408	379	209	44	3	2
	CDH74	13500	1	8	60	395	1166	2693	3764	3341	1649	384	33	6
	CDH80	5132	0	0	4	64	324	1048	1639	1399	566	85	3	0
(14) Wind	WSAvg	4.1	4.6	5.1	5.6	5.7	4.5	3.6	3.2	2.9	3.4	3.6	3.9	3.7
(15) Precipitation	PrecAvg	41.5	3.4	3.5	4.1	3.2	3.4	3.9	4.1	3.3	3.5	3.3	2.8	3.1
	PrecMax	60.0	7.2	8.0	8.4	10.7	7.2	10.0	10.1	7.9	10.5	14.0	5.3	6.5
	PrecMin	26.5	0.9	0.6	1.0	0.4	0.4	0.1	1.0	0.6	0.0	0.0	0.4	0.6
	PrecStd	6.7	1.7	1.7	1.9	1.9	1.7	2.3	2.1	1.8	2.3	2.8	1.4	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.6	74.9	80.0	84.4	90.0	94.6	96.8	98.8	92.8	88.0	78.9	72.6
		MCWB	59.7	61.1	59.3	64.1	70.1	73.0	72.7	73.5	69.6	71.4	62.6	62.9
	2%	DB	65.6	69.8	74.9	81.7	86.4	91.1	93.0	93.0	89.9	81.8	72.5	66.9
		MCWB	57.4	57.2	58.2	62.6	68.8	72.6	74.3	73.0	69.9	65.6	60.0	62.3
	5%	DB	62.7	64.4	71.9	79.1	83.8	89.6	90.8	90.3	86.2	78.7	69.7	63.5
		MCWB	55.1	54.6	57.2	61.7	68.4	72.1	74.2	73.9	70.7	66.0	59.3	57.5
	10%	DB	57.3	60.6	66.3	75.1	81.1	86.3	88.9	87.9	82.5	74.4	64.5	59.5
		MCWB	49.4	51.2	54.8	60.6	66.8	71.4	73.8	73.7	69.5	64.4	56.0	53.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	64.6	64.4	69.4	73.5	78.1	79.2	78.8	76.4	73.6	70.1	66.9
		MCDB	65.7	71.0	72.2	76.9	80.9	84.9	85.8	86.1	83.3	80.3	71.7	69.2
	2%	WB	60.0	61.4	62.3	66.7	72.0	76.2	77.8	77.5	74.9	71.9	64.6	63.4
		MCDB	64.2	65.7	68.8	74.7	80.3	84.8	86.8	86.0	82.1	78.2	68.3	66.4
	5%	WB	56.5	57.1	60.2	64.7	70.5	75.0	76.7	76.5	73.5	69.3	61.6	59.9
		MCDB	61.1	62.1	67.3	72.9	78.8	83.7	85.8	84.8	80.1	74.7	66.3	62.6
	10%	WB	51.5	52.3	57.7	63.0	69.1	73.8	75.5	75.5	72.2	66.8	58.2	54.6
		MCDB	56.4	58.2	64.9	71.1	77.0	82.0	84.1	83.4	78.6	72.5	64.2	58.1
(35) Mean Daily Temperature Range	5% DB	MDBR	19.8	20.4	22.1	23.0	21.9	20.5	19.4	18.9	19.9	21.9	23.3	20.0
		MCDBR	24.1	26.3	26.1	26.9	25.4	23.7	22.2	22.4	24.6	25.3	28.1	23.3
		MCWBR	16.4	16.5	13.4	11.6	10.1	8.3	7.6	7.3	8.8	12.2	16.8	16.2
		MCDBR	19.9	22.0	20.3	20.8	20.5	19.7	19.1	18.6	18.3	19.5	19.7	18.0
	5% WB	MCWBR	16.7	16.9	13.6	10.6	9.6	8.3	7.5	7.3	8.7	10.8	16.2	15.2
(40) Clear-Sky Solar Irradiance	taub		0.310	0.322	0.357	0.400	0.453	0.483	0.534	0.516	0.424	0.377	0.326	0.309
	taud		2.498	2.461	2.399	2.277	2.165	2.119	2.002	2.031	2.283	2.390	2.494	2.535
	Ebn at Noon		283	292	289	280	266	257	243	244	264	268	275	277
	Edh at Noon		26	30	35	42	48	50	56	53	39	32	26	24
(44) All-Sky Solar Radiation	RadAvg		820	1026	1366	1709	1901	2039	1902	1735	1459	1195	906	702
	RadStd		53	126	96	134	155	147	127	82	130	152	74	63

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+187

Nomenclature: See separate page